Zenith: testing repairs to an adaptive investigator

Will Cook • 13 October 2026

Abstract. Can an investigator improve its reusable procedure by testing a diagnosis of its own work? Zenith implements a small loop: Lab chooses numerical investigations, Oracle reviews the recorded computation against later observations, and Evolve proposes an exact procedure replacement. Code reruns the whole investigation under unchanged and candidate procedures and returns the measured difference to the next review. The first historical episode did not improve forecasts. A subsequent cheap-model audit separates elementary interpretation, output format, long-context review and task value. Modern models solve small supplied facts yet still make unsupported review claims. We repair cost controls and context duplication, and motivate an information-sufficient task with a useful learning signal. Neither execution nor calibration demonstrates learning.

1. Start with a competent investigator. A repair system needs something worth repairing. We supply a working investigator, accurate numerical tools, sensible references and an earlier worked example before evaluating model-authored changes. These are engineering choices, not discoveries attributed to the learning loop.

Lab is one adaptive model using three actions. It can *inspect* observations, *compare* a registered numerical representation, or *commit* forecasts. Each answer is available before the next action. A question must say what its possible answers would change. Lab can stop early, retain a reference or investigate across baskets; there is no compulsory question for each target.

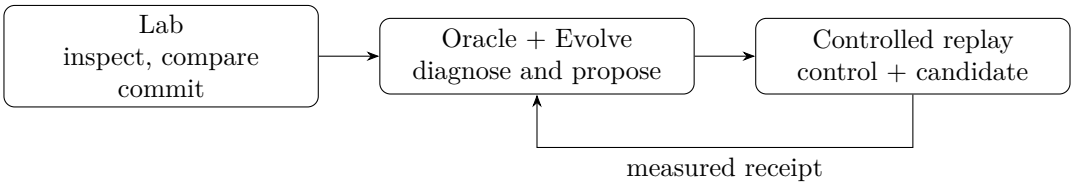
The retained stress test forecasts every basket's five-session return from the shared history of 147 eligible NEPSE instruments and 17 indices [1]. Six research baskets have membership fixed using earlier observations and fixed shares following equal initial-value allocation. For value V_b, the target is $r_{b,t} = V_b(t+5)/V_b(t) - 1$. These are constructed indices, not traded ETFs. Missing quotes retain their last mark and expose its age.

The tools preserve distinctions that matter: past returns versus future targets; member participation versus value contribution; predictive association versus accounting; and improvement over a weak comparator versus improvement over the selected reference. A distinction earns its place when it changes a useful question, decision or test. Requiring a long reasoning schema at every step is not itself a scientific contribution.

The numerical boundary. Code supplies ridge, zero-return and mature-training-mean references for each basket. Its advisory default minimises past validation error. An inspection returns constituent contributions and dated features. A comparison adds one summary—positive value weight, top-three absolute-contribution share or fresh quote weight—from any basket or the whole universe to the same ridge model. It keeps feature-count normalisation, regularisation, training rules and eight validation origins fixed, and reports paired results for all six targets.

The model chooses registered candidate IDs; code supplies their forecasts and target identities. Commitment requires all six targets. Invalid output stops the attempt without retry or a substitute forecast. Repeated searches over eight origins remain exploratory model selection. Price histories do not establish market causes.

2. One procedure, one measured repair. The native engine executes action/result pairs, with at most eight model turns and six comparisons. Early commitment makes the remaining slots inactive. The exported graph records the calls and supplied contexts that actually executed; its edges describe information flow, not economic causality. Public justifications and tool receipts are inspectable records, not access to the model's private internal reasoning.



Oracle diagnosis and Evolve's proposal share one model response. The reviewer receives the executed Lab graph, exact instructions and inputs, actions, numerical receipts and a separately labelled later outcome. A wrong forecast does not by itself locate an error. A proposed defect must cite an actual step, consider an alternative explanation and remain within available capabilities. No supported repair is a valid conclusion.

The editable object is one reusable investigation procedure of at most 4,096 characters. A candidate replaces it in full. The reviewer cannot silently alter the tools, model, targets or evaluator. Its proposal names an expected first action signature, an interpretation, a valid counterexample and a disconfirmer. The signature is a limited observable: an action occurring does not establish that the reasoning improved.

The coordinator freezes the procedure, snapshot, runtime, model and resource limits. It runs the complete Lab from entry with both the unchanged procedure and the exact proposed text. This tests changes to question selection as well as final choices. The receipt separates changed actions, numerical recipes, forecasts and losses. It retains failures. No candidate is automatically adopted.

The next actual reviewer request contains that exact receipt, verified in both the saved input and provider prompt. The reviewer must assess the tested proposal before revising, rejecting, retaining it for evaluation or making no change. Delivery is necessary for feedback use; it does not prove that the model understood or used it.

3. What would count as learning? Engineering fixes to tools or prompts first create a new baseline. Learning is then a model-authored procedure change evaluated within a frozen baseline. Keep these claims separate:

- **Execution:** the exact proposal ran and its measured effects reached the next review.
- **Behaviour:** the intended investigation or decision changed beyond ordinary repeated-run variation.
- **Predictive value:** the complete forecast vector improved under the declared loss and cost accounting.
- **Transfer:** a frozen reusable change helped on protected further cases, including counterexamples and regressions.

A same-case repair chosen after seeing the outcome supplies development evidence. One unchanged/candidate pair cannot establish a stochastic treatment effect. Numerical enumeration is an essential simpler comparator, with any larger search allowance disclosed. A multi-agent architecture must earn its extra cost against this adaptive controller using the same observations and tools.

4. Cost, competence and the remaining failure. On the exposed 2 February 2025 case, the original GPT-4.1 episode used 12 calls and cost $1.243572. It sent 635,022 input tokens: repeated full contexts made review requests unusually large. This model was explicitly selected; the receipts do not show overload forcing an expensive fallback.

Original, unchanged and proposed-procedure Lab runs all chose six zero-return forecasts, with MSE 0.000642668662. The reviewer falsely ranked a larger MSE below a smaller one and proposed an unnecessary minimum-error rule. A separately versioned correction cost $0.375084 and withdrew that diagnosis, but retained other unsupported claims. No repair was adopted.

New inference verifies cheap Chinese-developed models and endpoint tariffs, serialises requests, and stops without expensive fallback. A lossless evidence bank shrank the inspected subject from 2,115,068 to 258,745 bytes with exact request reconstruction. A capability inventory distinguishes the 21 legal recipes from those already tried.

Modern-model condition	Calls	Observed result
DeepSeek, original reviewer request	1	Correct ordering; unsupported proposal
DeepSeek, native Lab	8	Six zero forecasts; unchanged MSE
DeepSeek, lossless review	1	Misread remaining action space
Qwen, component calibration	1	16/16 semantic; wrong JSON shape
DeepSeek, component calibration	1	16/16; valid JSON shape
Qwen, capability-aware review	1	No change; false directional claim

These 13 calls cost $0.046824887272. Workloads and interfaces differ; this is not a controlled overall comparison. Models were DeepSeek V4.1 Flash and Qwen 3.8 Omni Flash. Two other attempts stopped at endpoint checks without inference. A Lab review exceeded the byte cap before a charge; its lossless continuation is separately versioned.

The calibration tests ranking, interval matching, reachable actions and mandatory versus advisory rules. Code solves it exactly; it establishes no LLM advantage. Qwen's malformed response remains a failure despite correct answers. Its later review called neutral forecasts directionally correct for falling baskets. No change is not proof of sound review.

Scope and next evidence. The larger question is what can accumulate across the loop: better concepts, better investigations, composable reasoning procedures, or an executable theory of an observed system. These are research possibilities, not measured results. The next design may replace the graph or task. It must explain a useful decision from available information and show a complete worked learning cycle. Ranking and forecast triage are examples, not the limit of the ambition.

February, May and August are exposed development cases. Shared conditions, overlapping horizons, possible pretraining knowledge and uncertified release vintages preclude trading-performance claims. DSPy [2], MIPRO [3] and TextGrad [4] already optimise model programmes or feedback. Novelty and practical advantage remain open. Source-bound cost, input and execution receipts accompany the repository.

References

[1] SocrateAI. *NEPSE Open Data*, pinned source `497e27238a10`. Dataset-specific MIT-form licence.

[2] O. Khattab et al. *DSPy*. 2023. arXiv:2310.03714.

[3] K. Opsahl-Ong et al. *Optimizing Instructions and Demonstrations for Multi-Stage Language Model Programs*. 2024. arXiv:2406.11695.

[4] M. Yuksekgonul et al. *TextGrad: Automatic Differentiation via Text*. 2024. arXiv:2406.07496.